

# Arevind Mohan

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## Professional Summary

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Associate Python Software Developer specializing in AI/ML and LLM-based application development. Hands-on experience building Retrieval-Augmented Generation pipelines, AI assistants, vector database workflows, document intelligence systems, and backend automation services. Skilled in Python, FastAPI, Qdrant, ChromaDB, Docker, Redis, Kafka, and production-oriented AI system design, with a strong focus on transforming unstructured data into reliable AI-powered applications.

## Technical Skills

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**Programming:** Python, C

**AI / ML / LLM Systems:** Retrieval-Augmented Generation (RAG), LLM Orchestration, Prompt Engineering, Agentic AI, LangChain, LangGraph, CrewAI, Ollama, Offline LLMs, Guardrails

**Vector Databases Retrieval:** Qdrant, ChromaDB, Embeddings, Semantic Search, Knowledge Base Preparation, Document Retrieval

**Backend Infrastructure:** FastAPI, Flask, Docker, Kafka, Redis, IMAP/SMTP Automation, REST APIs

**Data Processing:** Pandas, NumPy, PDF/Text Processing, Web Scraping, Structured Data Extraction

**Web Tools:** HTML, CSS, Git, GitHub, VS Code, Jupyter Notebook

## Professional Experience

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**Associate Software Engineer – Python / AI-ML**

**Aug 2025 – Present** *IonicX Digital*

- Building AI-powered applications by integrating LLMs, vector databases, retrieval pipelines, and backend services into enterprise products.
- Developed an internal **knowledge base preparation tool** that extracts structured information from websites, PDFs, and Q&A documents, enabling vector database creation for RAG-based applications.
- Built an **AI hospital assistant chatbot** that understands natural language symptom queries, maps them to relevant doctors, and supports appointment availability across WhatsApp, Facebook Messenger, and web interfaces.
- Developed a **CrewAI-based mall assistant chatbot** for a Dubai client, enabling users to discover movies by language, genre, and time, and locate stores through natural language queries over periodically scraped data.
- Working on an **AI-powered email assistant** for automated query handling using Qdrant-based RAG pipelines, IMAP/SMTP automation, offline multimodal LLMs, guardrails, Docker, Kafka, and Redis.
- Contributed to backend workflows for AI applications, including data ingestion, retrieval logic, prompt design, response validation, and deployment-oriented service integration.

## Projects

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**VecSeek – Document Retrieval Workspace**

- Building an AI-powered document retrieval workspace that allows users to upload files, organize them into folders, index them into vector databases, and query them through a clean UI or API.
- Designed the system to convert uploaded PDF, DOCX, and TXT files into folder-specific vector collections using embeddings and semantic search.
- Implemented document preprocessing workflows for extracting clean text, handling structured content, detecting duplicate files, and preparing metadata-rich chunks for retrieval.
- Developed retrieval testing features that allow users to select a folder, configure chunk settings, test queries, and inspect retrieved chunks before using the system in production workflows.
- Tech stack: Python, FastAPI, ChromaDB, embeddings, document parsing, vector search, REST APIs, and frontend integration.

## Education

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**B.Tech – Electronics and Communication Engineering 2020 – 2024** Mar Baselios College of Engineering and Technology CGPA: 6.83 / 10